

MUHAMMAD SALMAN KHALID

+44 (0) 7879 973806 | salmankhalid.data@gmail.com | [LinkedIn](#) | [GitHub](#) | London, UK | UK Passport

PROFILE

Economics graduate (Upper 2:1, Swansea) and UCL Business Analytics MSc who builds and validates machine learning models in Python - regression, classification and forecasting - on large, imperfect commercial datasets. Econometrics-trained, comfortable turning a business question into a testable hypothesis, and practised at explaining what a model does and does not support to non-technical stakeholders.

EDUCATION

University College London - MSc Business Analytics

Sep 2025 - Aug 2026

- Relevant modules: Predictive Analytics, Statistical Foundations of Business Analytics, Operations Analytics (pricing & segmentation), Data Engineering (SQL, Python), Natural Language Processing, Marketing Analytics, Business Analytics Consulting Project.

Swansea University - BSc Economics (Upper 2:1)

Sep 2021 - Sep 2024

- Core modules: Time series econometrics (forecasting), Microeconometrics & panel data models, Advanced macroeconomics, Advanced microeconomics.
- Vice President, Economics Society.

MACHINE LEARNING & ANALYTICS PROJECTS

Predictive Maintenance - Remaining Useful Life Modelling (UCL, Python)

- Built an end-to-end regression pipeline on 20,000+ sensor records across 100 run-to-failure histories; ran EDA across 21 sensors, removed 7 low-variance features and engineered 28 rolling-window features to capture degradation trends.
- Trained and compared linear regression, random forest and gradient boosting; shipped a tuned model at Test RMSE 18.43 and R^2 0.80, within the published benchmark range.
- Caught and corrected three material errors (data leakage, omitted baseline, implicit test-set exposure) via a plan - verify - revise loop with a written decision log, before they reached the result.
- Packaged the model with a written summary covering intended use, limitations and risks for non-technical readers.

Health Risk Classification - Supervised ML Pipeline (UCL, Python)

- Built a full supervised classification pipeline on a large public health dataset (BRFSS, ~250k records) predicting risk from demographic, behavioural and clinical features.
- Performed EDA and feature diagnostics (distributions, correlations, heatmaps); implemented categorical encoding, scaling and normalisation using Pandas and scikit-learn.
- Trained and compared logistic regression, decision tree and random forest models; evaluated on accuracy, precision, recall, F1, ROC curves and confusion matrices, with explicit analysis of which error type carried the greater cost.

Pricing Strategy & Revenue Optimisation - Consulting Case (UCL)

- Analytical lead on a six-person consulting team; translated a management brief on pricing into a quantified recommendation, presented as a client-style engagement.
- Modelled willingness-to-pay across a 10,000-member base and recommended a two-tier structure delivering +\$11,068 (+2.3%) revenue over the flat-rate optimum, with peak-hour congestion mitigations.
- Stress-tested the underlying survey - flagged hypothetical, selection and strategic-response bias - and recommended A/B price testing and a pilot before rollout rather than accepting the base case at face value.

Multi-Source Data Reconciliation & Risk Scoring (UCL, Python, SQL)

- Built an automated ETL pipeline reconciling 5 disparate sources (web scraping, Guardian API, Reddit API, Our World in Data, web archives) to test 10 FTSE 100 firms' published claims against independent evidence.
- Designed a composite risk-scoring framework and an analytical warehouse layer in DuckDB and Spark SQL serving 7+ analytical queries; implemented data quality controls including deduplication, cross-source reconciliation across 244 news articles and 664 social posts, and validation checks.
- Delivered as an interactive dashboard with 8 analytical views, from executive KPIs to per-company deep dives, so a non-technical reader could trace any figure back to its source.

WORK EXPERIENCE

SK Organisation (SKO) - Founder & Managing Director

Direct marketing firm serving major UK charities and brands

Sep 2024 - Aug 2025

Remote, UK

- Founded and ran the firm's P&L; led client (Natures Menu, BHF, TMM) conversations, scoped campaigns, and translated operational challenges into measurable workflows.
- Managed a team of 8; introduced performance metrics, forecasting and training that lifted conversions by 40%; embedded new tooling and KPIs on a weekly cycle.
- Awarded a Young Entrepreneur Award for leadership and innovation.

IEUK 2023 - Financial Services & Consulting Intern

Jul 2023 - Aug 2023

Bright Network

Remote

- Completed a PwC work-sample case assessing a company's financial records and internal processes; applied principles of materiality, evidence quality and decision-making under uncertainty.

SKILLS & INTERESTS

- **Languages & ML:** Python (Pandas, NumPy, scikit-learn, PyTorch), SQL, R, Spark.
- **Techniques:** Regression, classification, ensemble methods, time series forecasting, feature engineering, model evaluation and validation.
- **Tools:** Power BI, Tableau, Streamlit; Excel (pivot tables, lookups, financial modelling); Docker, Git.
- **Communication:** written logic trails, structured decision logs, client-style decks; explaining quantitative findings to non-technical stakeholders.
- **Languages:** English, Urdu, Hindi (native); Arabic (intermediate); Spanish (beginner).
- **Interests:** Cricket - ex-Berkshire County cricketer (5+ years); Founder and Captain, Reading Tigers CC (Division 5 to Premier Division), Senior Coach.